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 - 결론 및 주요 Feature
 - P10 · Hur 2025
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 - IMU 원시 신호 (8개 센서 × 9채널 = 72 variables)
 - IMU 부착 위치 (8개)
 - Target Variables (예측 대상 관절각, 10개)
 - 결론 및 주요 Feature

- P11 · Palazzo 2025
 - Input Variables
 - IMU 원시 신호 (5개 센서 × 9채널 = 45 channels)
 - IMU 부착 위치 (5개)
 - Target Variables (분류 클래스)
 - 결론 및 주요 Feature
- 전체 Input Variables 통합 비교표
- 중요 Feature 요약

Reference Papers — Input Variables 종합 분석표

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파일명 변경 내역

원본 파일명	변경 후 파일명
gait_biomechanics_in_anterior_cruciate.4.pdf	Goetschius_2018_gait_biomechanics_ACLR_time_frames.pdf
main.pdf	Ursei_2020_foot_ankle_compensation_ACL_deficiency_child
nihms917286.pdf	Erhart-Hledik_2018_longitudinal_knee_gait_mechanics_ACL
nihms-1713216.pdf	Krishnan_2022_mechanical_factors_knee_extension_moment
nihms-1826154.pdf	Garcia_2022_gait_asymmetries_walking_speed_ACLR.pdf
hwang-et-al-2024-machine-learning-models-for-predicting-return-to-sports-after-anterior-cruciate-ligament.pdf	Hwang_2024_ML_predicting_RTS_ACLR.pdf
fbioe-14-1762919.pdf	Yuan_2026_IMU_running_gait_ML_sensor_fusion.pdf
s41597-025-05959-w.pdf	Voisard_2025_clinical_gait_signals_wearable_IMU_dataset
s41597-025-06307-8 (1).pdf	Yona_2026_COMPWALK-ACL_multi-pace_IMU_gait_dataset.pdf
s41598-025-89716-4.pdf	Hur_2025_learning_lower_limb_kinematics_IMU.pdf
sensors-25-00260-v2.pdf	Palazzo_2025_deep_learning_pathological_gait_recognitio

P1 · Goetschius 2018

제목: Gait Biomechanics in Anterior Cruciate Ligament-reconstructed Knees at Different Time Frames Postsurgery 저자: John Goetschius, Jay Hertel, Susan A. Saliba, Stephen F. Brockmeier, Joseph M. Hart 저널: Medicine & Science in Sports & Exercise, Vol. 50, No. 11, pp. 2209–2216 연도: 2018 · DOI: 10.1249/MSS.0000000000001693 파일: [Goetschius_2018_gait_biomechanics_ACLR_time_frames.pdf](#)

Input Variables

변수	카테고리	단위	설명
External knee flexion moment	Kinetics (Sagittal)	$N \cdot m \cdot kg^{-1} \cdot m^{-1}$	슬관절 굴곡 외부 모멘트
External knee extension moment	Kinetics (Sagittal)	$N \cdot m \cdot kg^{-1} \cdot m^{-1}$	슬관절 신전 외부 모멘트
External knee adduction moment	Kinetics (Frontal)	$N \cdot m \cdot kg^{-1} \cdot m^{-1}$	슬관절 내전 외부 모멘트
External knee abduction moment	Kinetics (Frontal)	$N \cdot m \cdot kg^{-1} \cdot m^{-1}$	슬관절 외전 외부 모멘트
External hip adduction moment	Kinetics (Frontal)	$N \cdot m \cdot kg^{-1} \cdot m^{-1}$	고관절 내전 외부 모멘트
External hip abduction moment	Kinetics (Frontal)	$N \cdot m \cdot kg^{-1} \cdot m^{-1}$	고관절 외전 외부 모멘트
Knee flexion motion	Kinematics (Sagittal)	degrees	슬관절 굴곡 각도
Knee abduction motion	Kinematics (Frontal)	degrees	슬관절 외전 각도
Hip abduction motion	Kinematics (Frontal)	degrees	고관절 외전 각도
vGRF	Ground Reaction Force	$N \cdot kg^{-1}$	수직 지면반력 (체중 정규화)

원문 발췌 — 변수 정의 (Methods, "Data collection", p. 2210):

"Primary gait variables included knee and hip kinetics and kinematics in the sagittal and frontal planes and vertical ground reaction forces (vGRF)."

원문 발췌 — 단위 설명 (Methods, "Data processing", p. 2210):

"Vertical ground reaction forces were reported in newtons normalized by body mass ($N \cdot kg^{-1}$), kinematic variables were reported in rotational degrees, and kinetic variables were reported in external moments, newton-meters ($N \cdot m$) normalized to body mass and height ($N \cdot m \cdot kg^{-1} \cdot m^{-1}$)."

원문 발췌 — 측정 조건 (p. 2210):

"Walking and jogging trials were performed at standardized speeds of $1.34 \text{ m} \cdot \text{s}^{-1}$ (3.0 mph) and $2.68 \text{ m} \cdot \text{s}^{-1}$ (6.0 mph), respectively."

측정 장비: 12× Bonita-10 cameras (Vicon), split-belt instrumented treadmill (Bertec), Motion Monitor + Nexus software

결론 및 주요 Feature

원문 발췌 — 결과 요약 (Abstract - Results):

"Early ACLR group demonstrated reduced knee flexion, knee extension, knee adduction, and hip adduction moments on the ACLR limb. Mid ACLR group demonstrated no gait differences between limbs or other groups. Late ACLR group demonstrated reduced knee flexion moments, and greater knee and hip adduction moments in their ACLR limb."

원문 발췌 — "Quadriceps Avoidance" 패턴 (Discussion, p. 2214):

"The presence of the 'quadriceps avoidance' pattern in the late ACLR group may be related to an inability of the quadriceps muscles to eccentrically absorb joint loads. Theoretically, an inability to adequately attenuate forces on the knee could lead to progressive degradation of articular surfaces."

원문 발췌 — 결론 (Conclusions, p. 2215):

"The early ACLR group demonstrated lower sagittal and frontal plane joint loading on the ACLR limb compared with contralateral and control limbs. The late ACLR group demonstrated lower sagittal plane joint loading compared with control limbs and greater frontal plane joint loading compared with contralateral and control limbs."

핵심 중요 변수: vGRF, external knee extension moment, knee flexion motion, external knee adduction moment, hip abduction motion (Early ACLR Walking에서 유의한 집단 내 비대칭 차이 확인)

P2 · Ursei 2020

제목: Foot and ankle compensation for anterior cruciate ligament deficiency during gait in children **저**

자: Monica E. Ursei, Franck Accadbled, Marino Scandella, Gorka Knorr, Caroline Munzer, Pascal

Swider, Jérôme Briot, Jérôme Sales de Gauzy **저널:** Orthopaedics & Traumatology: Surgery &

Research, Vol. 106, pp. 179–183 **연도:** 2020 · DOI: 10.1016/j.otsr.2019.07.009 **파일:**

[Ursei_2020_foot_ankle_compensation_ACL_deficiency_children.pdf](#)

Input Variables

변수	카테고리	측정 시점 (Gait Cycle %)	단위
Ankle dorsiflexion / plantar flexion	Kinematics (Ankle)	0%, 25%, 60%, 83% GC	degrees
External foot progression angle	Kinematics (Foot)	25% (stance), 70% (swing)	degrees
Walking speed	Spatiotemporal	—	m/s
Cadence	Spatiotemporal	—	steps/min
Stride length	Spatiotemporal	—	m
Step length	Spatiotemporal	—	m
Single-leg stance (%)	Spatiotemporal	—	% of GC
Double-leg stance (%)	Spatiotemporal	—	% of GC

원문 발췌 — 측정 변수 정의 (Methods, p. 180):

"The following spatiotemporal parameters were recorded: speed, cadence, stride and step length, and percentages of time spent in single-leg and double-leg stance. Kinematic data on the ankle and foot in the sagittal plane were compared to those obtained in a control group of 37 healthy children."

원문 발췌 — 발목 각도 측정 시점 (p. 180):

"Ankle flexion and extension were computed at the following points of the GC: initial contact (0%), mid-stance (25%), terminal stance (60%), and swing phase (83%). The foot progression angle was determined at mid-stance (25%) and during the swing phase (70%)."

측정 장비: Vicon 360 (5× 60-Hz IR cameras), 15 reflective markers (Helen Hayes model)

결론 및 주요 Feature

주요 결과 (Abstract — Results):

"Compared to the reference values, the ankle was in plantar flexion at initial contact in 41 patients, and ankle dorsiflexion during the stance phase was diminished in 39 patients."

수치 결과 (Table 1, p. 181):

변수	환자군 (n=47)	대조군 (n=37)	p값
Ankle — 0% GC (Initial contact)	-3.44 ± 3.54°	0.74 ± 3.67°	2.55×10 ⁻⁹
Ankle — 25% GC (Mid-stance)	5.91 ± 2.15°	7.96 ± 2.52°	1.28×10 ⁻¹¹
Ankle — 60% GC (Terminal stance)	-14.18 ± 6.40°	-12.99 ± 6.60°	>0.05
Ankle — 83% GC (Swing)	0.78 ± 2.72°	1.09 ± 3.43°	>0.05

원문 발췌 — 결론 (p. 182):

"Our findings confirm the hypothesis that children alter their gait to compensate for ACL deficiency. In particular, plantar flexion of the ankle at initial contact was noted. Compensatory behaviours during gait may be difficult to detect during the physical examination. Gait analysis is a reliable tool for elucidating compensatory mechanisms and optimising rehabilitation therapy."

핵심 중요 변수: Ankle dorsiflexion at initial contact (0% GC), ankle dorsiflexion at mid-stance (25% GC) — 통계적으로 가장 유의한 ACL 결핍 지표

P3 · Erhart-Hledik 2018

제목: Longitudinal changes in knee gait mechanics between 2 and 8 years after anterior cruciate ligament reconstruction 저자: Jennifer C. Erhart-Hledik, Constance R. Chu, Jessica L. Asay, Thomas P. Andriacchi 저널: Journal of Orthopaedic Research, Vol. 36, No. 5, pp. 1478-1486 연도: 2018 · DOI: 10.1002/jor.23770 파일: [Erhart-Hledik_2018_longitudinal_knee_gait_mechanics_ACLR.pdf](#)

Input Variables

관절 Kinetics (역학적 변수)

변수	약어	설명
Peak knee flexion moment	KFM	최대 슬관절 굴곡 모멘트
Knee adduction moment (1st peak)	KAM1	슬관절 내전 모멘트 1차 피크
Knee adduction moment (2nd peak)	KAM2	슬관절 내전 모멘트 2차 피크
Knee adduction moment impulse	KAM Impulse	슬관절 내전 모멘트 총격량
Knee extension moment (1st peak)	KEM1	슬관절 신전 모멘트 1차 피크
Knee extension moment (2nd peak)	KEM2	슬관절 신전 모멘트 2차 피크
Peak internal rotation moment	KIRM	슬관절 내회전 모멘트
Peak external rotation moment	KERM	슬관절 외회전 모멘트

원문 발췌 — Kinetics 변수 정의 (Gait Analysis, p. 4):

"For kinetic analyses, our outcome variables included the peak knee flexion moment (KFM), knee adduction moment first peak (KAM1), second peak (KAM2), and impulse (KAM Impulse),

first and second peak extension moments (KEM1 and KEM2), and peak internal (KIRM) and external (KERM) rotation moments."

관절 Kinematics (운동학적 변수)

변수	측정 구간	설명
Peak knee flexion angle	Stance 전반부	스탠스 전반부 최대 굴곡 각
Minimum knee flexion angle	Stance 후반부	스탠스 후반부 최소 굴곡 각
Peak varus angle	Stance 전반부	최대 내반 각도
Peak valgus angle	Stance 후반부	최대 외반 각도
Average external rotation angle	Whole stance (heel-strike → toe-off)	평균 외회전각
Average anterior femoral displacement	Whole stance	평균 대퇴골 전방 변위

원문 발췌 — Kinematics 변수 정의 (Gait Analysis, p. 4-5):

"For kinematic analyses, our outcome variables included the peak knee flexion angle during the first half of stance phase, minimum knee flexion angle in the second half of stance phase, peak varus angle in the first half of stance phase, peak valgus angle in the second half of stance phase, average external rotation angle during stance (heel-strike to toe-off), and average anterior femoral displacement during stance (heel-strike to toe-off)."

Spatiotemporal Variables

변수	설명
Walking speed	보행 속도
Step length	한 걸음 길이

측정 장비: Qualisys Medical optoelectronic system (120 Hz), Bertec force plate, BioMove software (Stanford), Cardan/joint coordinate system 기반 3D kinematics

결론 및 주요 Feature

원문 발췌 — Kinetics 결과 (Results - Kinetics, p. 5):

"Over the follow-up period, there was a reduction in the magnitude of KEM1 ($p=0.048$) and KEM2 ($p<0.001$), while an increase in peak KFM ($p=0.002$) was seen. KAM1 ($p=0.009$), KAM2 ($p=0.009$), KAM Impulse ($p=0.004$), and KIRM ($p<0.001$) decreased over follow-up."

원문 발췌 — Kinematics 결과 (Results - Kinematics, p. 6):

"Over the follow-up period, there was a significant increase in peak knee flexion angle in the first half of stance phase ($p=0.026$) and in the minimum knee flexion angle in the second half of stance phase ($p<0.001$). The average external rotation angle increased from 2 to 8 years post-ACLR ($p=0.007$), while a reduction in average anterior femoral displacement was observed ($p=0.006$)."

원문 발췌 — Discussion (p. 6):

"The results of this study demonstrate that there are longitudinal changes from 2 to 8 years post-ACLR in knee joint kinetic and kinematic features that have been related to clinical patient-reported outcomes after ACLR and the progression of knee OA."

종단 변화 방향 요약:

변수	2→8년 변화	p값
KFM	증가 (호전)	0.002
KEM1	감소	0.048
KEM2	감소	<0.001
KAM1	감소	0.009
KIRM	감소	<0.001
Peak flexion angle (1st half stance)	증가 (호전)	0.026
External rotation angle	증가 (악화 가능)	0.007

P4 · Krishnan 2022

제목: Mechanical Factors Contributing to Altered Knee Extension Moment during Gait after ACL Reconstruction: A Longitudinal Analysis
저자: Chandramouli Krishnan, Alexa K. Johnson, Riann M. Palmieri-Smith
저널: Medicine & Science in Sports & Exercise, Vol. 54, No. 12, pp. 2208–2215
연도: 2022 · DOI: 10.1249/MSS.0000000000003014
파일: [Krishnan_2022_mechanical_factors_knee_extension_moment_ACLR.pdf](#)

Input Variables (Predictors)

변수	카테고리	측정 구간	단위
Knee flexion angle at initial contact	Kinematics	Initial contact	degrees
Peak knee flexion angle	Kinematics	Early stance (first 50%)	degrees
Peak vGRF	Kinetics	Early stance	% body weight

원문 발췌 — Predictor 정의 (Abstract / Methods, p. 1):

"knee flexion angle at initial contact, peak knee flexion angle, and vertical ground reaction force (vGRF) contribute to knee extension moments during gait in individuals with anterior cruciate ligament (ACL) reconstruction."

원문 발췌 — 변수 측정 (Gait Biomechanics Testing, p. 4):

"Knee flexion angle at initial contact, peak knee flexion angle, peak vGRF, and peak knee extension moment were calculated for each limb during the early stance phase of gait for all three timepoints."

원문 발췌 — vGRF 정규화 (p. 4):

"vGRF was normalized to the participant's body weight. All biomechanical data were time normalized to the stance phase of the gait cycle."

원문 발췌 — 역학적 중요성 근거 (Introduction, p. 2):

"mechanical factors such as knee flexion angle and vertical ground reaction forces (vGRFs) could directly influence knee joint loading, as they serve as primary inputs to the calculation of knee joint moments and contact forces."

Output Variable (Target)

변수	설명
Peak knee extension moment	최대 슬관절 신전 모멘트 (Early stance)

원문 발췌 — 회귀 모델 정의 (Statistical Analyses, p. 4):

"Stepwise multiple linear regression analysis was used to determine the contribution of knee flexion angle at initial contact, peak knee flexion angle during early stance, and peak vGRF during early stance to knee joint loading (i.e., peak knee moment) during the early stance phase of gait."

측정 장비: 15-camera MX-13 Vicon (240 Hz), AMTI OR6-7 force plates (1200 Hz), 34개 14mm 반사 마커, Visual 3D software (v3.90)

결론 및 주요 Feature

원문 발췌 — 설명력 (Discussion, p. 6):

"these variables explained more than 75% of the variance in the peak knee extension moment data in both the ACL-reconstructed and the non-ACL-reconstructed limbs... peak knee flexion angle and peak vGRF symmetry values still explained more than 67% of the variance in peak knee extension moment symmetry."

원문 발췌 — 결론 (Abstract Conclusions, p. 2):

"Standardized beta coefficients indicated that changes in knee flexion angle had a greater impact (>2x) on knee extension moments than vGRF at both timepoints in both limbs ($\beta_{vGRF} = 0.204-0.309$; $\beta_{kneeflexion} = 0.703-0.831$)."

회귀 모델 성능 요약:

분석 조건	R ²	BF ₁₀
Reconstructed limb (raw)	0.767	>1000
Non-reconstructed limb (raw)	0.815	>1000
Symmetry values	0.673	>1000
Change scores (TP2-TP1, ACL)	0.775	>1000
Change scores (TP3-TP1, non-ACL)	0.883	>1000

핵심 중요 변수: Peak knee flexion angle ($\beta = 0.703-0.831$, vGRF의 2배 이상 영향력)

P5 · Garcia 2022

제목: Gait Asymmetries are Exacerbated at Faster Walking Speeds in Individuals with Acute Anterior Cruciate Ligament Reconstruction
저자: Steven A. Garcia, Scott R. Brown, Mary Koje, Chandramouli Krishnan, Riann M. Palmieri-Smith
저널: Journal of Orthopaedic Research, Vol. 40, No. 1, pp. 219-230
연도: 2022 · DOI: 10.1002/jor.25117
파일: [Garcia_2022_gait_asymmetries_walking_speed_ACLR.pdf](#)

Input Variables

변수	카테고리	설명	단위
Vertical GRF (vGRF)	Ground Reaction Force	수직 지면반력 파형	% body weight
Posterior-Anterior GRF (PA-GRF)	Ground Reaction Force	전후방향 지면반력 파형	% body weight
GRF Asymmetry (vGRF)	Derived	Injured limb GRF – Uninjured limb GRF	% body weight

변수	카테고리	설명	단위
GRF Asymmetry (PA-GRF)	Derived	Injured limb GRF – Uninjured limb GRF	% body weight
Walking speed condition	Experimental factor	80%, 100%, 120% self-selected speed	m/s

원문 발췌 — 측정 변수 (Abstract, p. 1):

"Bilateral vertical and posterior-anterior GRFs were recorded at each speed."

원문 발췌 — GRF 정규화 (Data Reduction, p. 5):

"Ensemble GRF curves were normalized to percent body weight (% BW) by dividing each element of the GRF waveform by the participant's body weight in Newtons and multiplying by 100."

원문 발췌 — 비대칭 산출 방식 (Data Reduction, p. 5):

"For ACL-reconstructed participants, the uninjured limb GRF was subtracted from the injured limb GRF across each percent of the stance phase."

원문 발췌 — PA-GRF 연구 동기 (Introduction, p. 3):

"evaluation of other components of the GRF during walking, such as the posterior-anterior GRF, have received far less attention in the literature despite evidence suggesting this parameter may provide additional insight into the extent of biomechanical asymmetries present after surgery."

측정 장비: Bertec split-belt treadmill (ITC-11-20L/R), LabVIEW 2011, Butterworth filter (8th order, 500 Hz lowpass) 통계: Statistical Parametric Mapping (SPM1D), 2×3 repeated-measures ANOVA (group × speed)

결론 및 주요 Feature

원문 발췌 — 주요 결과 (Discussion, p. 7):

"We found that the asymmetry in both vertical and posterior-anterior GRFs increased at faster walking speeds but decreased at slower speeds in individuals with ACL reconstruction. Conversely, speed did not affect GRF asymmetries in uninjured controls."

원문 발췌 — 결론 (Conclusion, p. 10):

"Increased walking speed magnifies biomechanical asymmetries in the vertical and posterior-anterior GRF in individuals with ACL reconstruction. At slower speeds, individuals with ACL reconstruction can walk with more symmetrical GRFs (i.e., decreased interlimb differences) compared with self-selected and fast speeds."

수치 결과:

조건	vGRF 비대칭	통계
80% 선호 속도	~10% BW	유의
120% 선호 속도	~18~20% BW	유의
ACL군 Group×Speed interaction	—	F*[2,86]=7.07, p<0.05
정상 대조군	속도 간 차이 없음	p>0.05

핵심 중요 변수: vGRF asymmetry, PA-GRF asymmetry — 속도가 증가할수록 비대칭 악화

제목: Machine learning models for predicting return to sports after anterior cruciate ligament reconstruction: Physical performance in early rehabilitation 저자: Ui-jae Hwang, Jin-seong Kim, Keong-yoon Kim, Kyu-sung Chung 저널: DIGITAL HEALTH (Sage) 연도: 2024 · DOI: 10.1177/20552076241299065 파일: Hwang_2024_ML_predicting_RTS_ACLR.pdf

Input Variables (수술 후 3개월 측정, 8개 변수)

변수	측정 도구	카테고리
Age	인구통계	Demographic
BMI	인구통계	Demographic
60°/s knee extensor peak torque (PT)	HUMAC-NORM isokinetic	Muscle strength
60°/s knee flexor peak torque (PT)	HUMAC-NORM isokinetic	Muscle strength
180°/s knee extensor average power (AP)	HUMAC-NORM isokinetic	Muscle power
180°/s knee flexor average power (AP)	HUMAC-NORM isokinetic	Muscle power
YBT total score	Y-Balance Test	Dynamic balance
BBS overall index	Biodex Balance System	Static/dynamic balance

원문 발췌 — 예측 변수 목록 (Pre-processing section, p. 4):

"This study included eight numerical predictors (age, body mass index [BMI], 60°/s knee extensor PT, 60°/s knee flexor PT, 180°/s knee extensor AP, 180°/s knee flexor AP, YBT total score, and BBS overall index)."

원문 발췌 — Biodex Balance System 설명 (Methods, "Biodex balance system test", p. 2):

"Postural stability was measured using the BBS (Biodex Medical Systems Inc., Shirley, NY, USA). The BBS electronically generates the anteroposterior (sagittal plane), mediolateral (frontal plane), and overall indexes. This study's overall index included the anteroposterior (sagittal plane) and mediolateral (frontal plane) indexes in detail."

원문 발췌 — Y-Balance Test 설명 (Methods, "Y-balance test", p. 3):

"Dynamic balance was measured using the YBT (Move2Perform; Evansville, IN, USA). The patients were instructed to stand on their weight-bearing leg in a box and gently push the side of the box using the unsupported leg, reaching as far as possible in the anterior, posteromedial, and posterolateral directions. The average distances achieved in the three directions were analyzed to assess dynamic balance."

원문 발췌 — 등속성 근력 설명 (Methods, "Isokinetic muscle strength test", p. 3):

"Isokinetic muscle strength was measured using the HUMAC-NORM isokinetic extremity system... The peak torque (PT) and the average power (AP) of the knee flexors and extensors were assessed at the 60°/s and 180°/s angular velocities, respectively."

Target Variables (수술 후 12개월 측정, RTS 결과)

변수	설명
Single-leg hop test	편측 홉 거리
Single-leg vertical jump test	편측 수직 점프 높이
Tegner activity score	스포츠 활동 수준 점수

결론 및 주요 Feature

모델 성능 요약:

Target	최고 모델	AUC (Test)	Accuracy	F1
Single-leg hop test	Random Forest	0.952	0.850	0.880
Tegner activity score	Random Forest	0.949	0.950	0.952
Single-leg vertical jump	Gradient Boosting	0.868	—	—

원문 발췌 — 주요 예측 변수 (Results, p. 5):

"RTS-single leg hop test was best predicted by random forest, with YBT total score, BMI, 60°/s knee extensor PT and 180°/s knee extensor AP in feature importance, and high 60°/s knee extensor PT, high YBT total score, high 180°/s knee extensor AP, low BMI in Shapley additive explanation being key factors."

원문 발췌 — 결론 (Conclusion, p. 9):

"The findings highlighted modifiable factors such as 60°/s knee extensor PT, YBT total score, 180°/s knee extensor and flexor AP, BBS overall index, and BMI in predicting successful RTS."

핵심 중요 변수 (SHAP 기반): YBT total score, 60°/s knee extensor PT, 180°/s knee extensor AP, BMI (낮을수록 RTS 성공 예측)

P7 · Yuan 2026

제목: Optimizing wearable IMU configurations for running gait analysis: a machine learning-based sensor fusion approach 저자: Ye Yuan, Yaohui Yu, Shanshan Cai, Weidong Cheng 저널: Frontiers in Bioengineering and Biotechnology 연도: 2026 · DOI: 10.3389/fbioe.2026.1762919 파일: [Yuan_2026_IMU_running_gait_ML_sensor_fusion.pdf](#)

Input Variables

센서 배치 조건 (독립변수)

구성	위치	IMU 수
Config 1 (C1)	Lumbar (L5/S1) only	1
Config 2 (C2)	Bilateral Ankles only	2
Config 3 (C3)	Lumbar + Bilateral Ankles	3

원문 발췌 — 구성 정의 (Methods, "Minimal configuration subset construction", p. 3):

"We tested three primary minimal configurations: Config 1 (C1): Lumbar-Only, using only the L5/S1 sensor (1 IMU); Config 2 (C2): Ankles-Only, using both ankle sensors (2 IMUs); and Config 3 (C3): Lumbar + Ankles, combining the L5/S1 and both ankle sensors (3 IMUs)."

원시 신호 입력

원문 발췌 — Raw signal (Methods, "Feature engineering", p. 3):

"Raw IMU signals (3-axis acceleration and 3-axis angular velocity) were processed using a sliding-window approach to transform high-frequency time-series data into a feature space suitable for regression."

추출된 Feature 목록 (Time-domain + Frequency-domain)

원문 발췌 — Feature 추출 (Methods, "Feature engineering", p. 4):

"Time-domain statistics, including the mean, standard deviation (STD), root mean square (RMS), minimum, maximum, peak-to-peak amplitude, skewness, kurtosis, and zero-crossing rate, were

calculated to capture the signal intensity and morphological characteristics of the impact and swing phases. Complementing these, frequency-domain features—specifically dominant frequency, spectral energy, and spectral entropy—were extracted after applying a Fast Fourier Transform (FFT)."

도메인	Feature	설명
Time	Mean	평균
Time	STD	표준편차
Time	RMS	제곱평균제곱근
Time	Min / Max	최솟값 / 최댓값
Time	Peak-to-peak amplitude	피크-피크 진폭
Time	Skewness	왜도
Time	Kurtosis	첨도
Time	Zero-crossing rate	영점 교차율
Frequency	Dominant frequency	지배 주파수
Frequency	Spectral energy	스펙트럼 에너지
Frequency	Spectral entropy	스펙트럼 엔트로피

적용 축: 가속도(Acc) X/Y/Z, 각속도(Gyro) X/Y/Z → sensor 위치별 조합

Target Variables (예측 대상)

원문 발췌 — Target 정의 (Table 2, "Definitions of target running gait parameters", p. 4):

파라미터	단위	정의
Cadence	steps/min	Steps per minute
Ground Contact Time (GCT)	ms	Duration from initial foot-strike (IC) to toe-off (TO)
Flight Time (FT)	ms	Duration from one foot's TO to the other foot's IC
Vertical Oscillation (VO)	cm	Peak-to-trough vertical displacement of L5/S1 during gait cycle
Gait Symmetry Index (SI)	%	Percentage difference between left and right GCT

결론 및 주요 Feature

Top 10 Features (GCT 예측, Config 1 기준, RFE 적용, Table 5, p. 8):

Rank	Feature	Axis	Type	Gini Importance
1	Accel_Z_variance	Z	Accel	0.187
2	Accel_X_variance	X	Accel	0.112
3	Accel_Z_rms	Z	Accel	0.091
4	Gyro_Y_energy_0-5Hz	Y	Gyro	0.075
5	Accel_Z_mean	Z	Accel	0.066
6	Accel_X_rms	X	Accel	0.051
7	Gyro_Y_variance	Y	Gyro	0.040
8	Accel_Z_kurtosis	Z	Accel	0.032
9	Accel_X_mean	X	Accel	0.029

Rank	Feature	Axis	Type	Gini Importance
10	Accel_Y_variance	Y	Accel	0.025

모델 성능 (Table 3, p. 6):

Target	Config	R ²	RMSE	MAPE
Cadence	C1 (Lumbar)	0.99	1.15 spm	0.85%
Vertical Oscillation	C1	0.96	0.41 cm	4.12%
Ground Contact Time	C1	0.95	7.98 ms	4.88%
Flight Time	C1	0.91	10.12 ms	6.15%
Gait Symmetry Index	C1	0.52	4.55	21.45%
Gait Symmetry Index	C3	0.91	1.90	7.12%

원문 발췌 — 결론 (Conclusion, p. 10):

"We confirmed that while a single lumbosacral IMU is surprisingly powerful, it is blind to asymmetry. We identified a three-sensor configuration (lumbosacral + bilateral ankles) as the minimal-optimal solution, capable of accurately predicting a comprehensive suite of global, temporal, and symmetry-based running gait parameters ($R^2 > 0.91$; MAPE $< 8\%$ for all)."

원문 발췌 — Feature 중요성 해석 (Discussion, p. 8):

"The dynamics of the CoM, particularly its vertical and anteroposterior acceleration (which our feature importance analysis confirmed as critical), are a direct reflection of the forces produced by and acting upon the lower limbs."

핵심 중요 변수: **Accel_Z_variance** (Lumbar, Gini 0.187), **Accel_X_variance** — 수직·전후 가속도 분산이 GCT 예측의 핵심

P8 · Voisard 2025

제목: A Dataset of Clinical Gait Signals with Wearable Sensors from Healthy, Neurological, and Orthopedic Cohorts
저자: Cyril Voisard, Rémi Barrois, Nicolas de l'Escalopier, Nicolas Vayatis, Pierre-Paul Vidal, Alain Yelnik, Damien Ricard, Laurent Oudre
저널: Scientific Data (Nature Portfolio), Vol. 12, Article 1674
연도: 2025 · DOI: 10.1038/s41597-025-05959-w
파일:

[Voisard_2025_clinical_gait_signals_wearable_IMU_dataset.pdf](#)

Input Variables

Raw IMU 신호 변수 (Table 3, p. 7)

변수명	설명	단위
Acc_X	Acceleration, x-axis	m/s ²
Acc_Y	Acceleration, y-axis	m/s ²
Acc_Z	Acceleration, z-axis	m/s ²
Gyr_X	Angular velocity, x-axis	rad/s
Gyr_Y	Angular velocity, y-axis	rad/s
Gyr_Z	Angular velocity, z-axis	rad/s

원문 발췌 — Raw signal 설명 (Table 3, p. 6-7):

"The main data labels for both sensors are summarized in Table 3, with corresponding units."

IMU 부착 위치: HE (머리), LB (요추 L4/L5), LF (왼발 등쪽), RF (오른발 등쪽)

원문 발췌 — 센서 위치 (p. 2-3):

"Four IMU devices were attached to the head (HE), lower back L4/L5 (LB), and on the dorsal face of each foot (LF for the left foot and RF for the right foot) of the participants."

변수 (센서별)	설명
{HE/LB/LF/RF}_Acc_X/Y/Z	원시 가속도 3축
{HE/LB/LF/RF}_FreeAcc_X/Y/Z	중력 성분 제거 가속도 3축
{HE/LB/LF/RF}_Gyr_X/Y/Z	각속도 3축

원문 발췌 — 전처리 변수 설명 (Table 4, p. 7):

"Main time-series labels for preprocessed data. The axes are given in the sensors reference frame."

메타데이터 (임상/인구통계) 변수 (Table 5, p. 8-9)

원문 발췌 — Metadata 설명 (p. 8):

"The first metadata are specific to the patient (epidemiological and demographic data) at the time of registration. They include the subject label, trial numbering, age, gender, height, weight, body mass index, laterality, group, pathology and clinical or radioclinical score."

Key	설명
age, gender, height, weight, BMI	인구통계
pathology, group	질환군 분류
evaluationScoreName / Value	코호트별 임상 점수
TUG	Time-Up and Go test (초)
visualGaitEvaluation	의사 보행 시각 평가 (0-4점)
leftGaitEvents / rightGaitEvents	Toe-off, Heel-strike 샘플 위치
uturnBoundaries	U턴 구간 경계

코호트별 임상 점수 (Table 2)

코호트	질환	임상 점수
HS	정상	—
HOA	고관절 골관절염	WOMAC (/100)
KOA	슬관절 골관절염	WOMAC (/100)
ACL	ACL 재건	IKDC (/100)
CVA	뇌졸중	FMA-LE (/34)
PD	파킨슨병	UPDRS III (/108)
CIPN	화학요법 유발 신경병증	TNSc (/28)
RIL	하지 방사선	OCS (/30)

파생 시공간 보행 파라미터 (Table 10)

변수	약어	단위
Gait velocity	V	m/s
Step length	SteL	m
U-turn time	UtrT	s
Stride time	StrT	s
CV of stride time	CVStrT	%
Double stance time proportion	dstT	%
CV of double stance time	CVdstT	%

측정 장비: XSens™ (± 2000 deg/s, 100 Hz), Technoconcept® I4 Motion (100 Hz); Butterworth LPF (8th order, 14 Hz cutoff)

결론 및 주요 Feature

원문 발췌 — 데이터 품질 검증 (p. 11):

"Statistically, the healthy cohort's results align closely with normative literature values, showing no significant differences compared to published norms ($p > 0.05$, independent two-sample t-tests)."

데이터 규모: 260명, 1,356개 보행 시험, 총 11시간 이상, 9,492개 파일 **Gait event 탐지 정확도:** Technoconcept 99.2%, XSens 99.1%

코호트별 보행 속도:

코호트	Gait speed (m/s)	Step length (m)
HS (Healthy)	1.17 (0.22)	0.67 (0.10)
ACL	0.98 (0.24)	0.62 (0.10)
KOA	0.76 (0.21)	0.51 (0.12)

P9 · Yona 2026

제목: The COMPWALK-ACL: A Dataset of Multi-pace IMU Gait Kinematics in Adolescents, Adults, and ACL Injured Patients **저자:** Tomer Yona, Bezalel Peskin, Arielle Fischer **저널:** Scientific Data (Nature Portfolio) **연도:** 2026 · DOI: 10.1038/s41597-025-06307-8 **파일:** [Yona_2026_COMPWALK-ACL_multi-pace_IMU_gait_dataset.pdf](#)

Input Variables

시공간 보행 파라미터

원문 발췌 — 신뢰도 평가 변수 (Technical Validation, p. 4):

"Within-participant reliability was assessed by calculating the coefficient of variation (CV) for five spatiotemporal gait parameters: gait speed, cadence, stride length, step width, and stride/step time."

변수	단위
Gait speed	m/s
Cadence	steps/min
Stride length	m

변수	단위
Step width	cm
Stride/Step time	s

하지 관절 키네마틱스 (Lower Limb Joint Kinematics)

원문 발췌 — 데이터 구조 (Data Records, p. 3):

"The .xlsx file contains a flattened, tabular version of key time-series variables... 'Segment Orientation', provided in both Quaternion and Euler angle formats for rotational data; 'Segment Position', detailing the 3D coordinates of each segment; and sheets for 'Segment Velocity', 'Segment Acceleration', 'Segment Angular Velocity', and 'Segment Angular Acceleration'."

원문 발췌 — 관절각 변수 (Data Records, p. 3):

"There are sheets for 'Joint Angles' and 'Ergonomic Joint Angles'... clear anatomical labels for joint angle variables, such as 'Left Ankle Dorsiflexion/Plantarflexion'... Furthermore, the data includes a sheet for the 'Center of Mass', which provides the position, velocity, and acceleration of the body's overall center of mass."

포함 관절각 변수:

관절	면	변수명 예시
Hip	Sagittal	Hip Flexion/Extension
Knee	Sagittal	Knee Flexion/Extension
Ankle	Sagittal	Ankle Dorsiflexion/Plantarflexion

Raw IMU 신호 변수

원문 발췌 — 센서 사양 (Equipment and calibration, p. 3):

"we recorded lower limb kinematics utilizing a model that consists of seven lightweight (16 grams) sensors, each containing a 3D accelerometer, a 3D gyroscope, and a 3D magnetometer. The recordings were captured at a frequency of 100 Hz."

센서 부착 위치 (7개): 양쪽 발, 양쪽 경골, 양쪽 대퇴, 골반

신호 종류	설명
3D 가속도 (free acceleration)	중력 보정 가속도
3D 각속도 (angular velocity)	회전 속도
3D 자기장 (magnetic field)	방향 추정 보조
Segment 위치 / 속도 / 가속도	세그먼트별 3D 좌표
Quaternion / Euler angle	방향 (orientation)
Heel/Toe contact	이진 발 접지 이벤트

메타데이터 (ID.csv)

원문 발췌 — 메타데이터 키 (Data Records, p. 4):

"a metadata file titled ID.csv provides participant information, including: ID, Group, Sex: Male/Female, Age: in years, Mass: Body mass in kilograms, Height: Stature in centimeters, Injured leg: Side of injury (for ACL participants only)"

결론 및 주요 Feature

원문 발췌 — 집단 간 차이 (Cohort Comparisons by Speed, p. 5):

"At normal walking speed, both ACLD and ACLR groups demonstrated significantly slower gait speed than healthy adults (ACLD: 1.19 ± 0.01 m/s; ACLR: 1.19 ± 0.02 m/s; healthy adults: 1.33 ± 0.02 m/s; both $p < 0.001$). Cadence was significantly lower in both ACL groups relative to healthy adults."

코호트별 보행 파라미터 (Table 4):

Cohort	Gait Speed (m/s)	Cadence (steps/min)	Stride Length (m)	Stride Time (s)
Healthy Adults	1.37 ± 0.47	112 ± 21.60	1.44 ± 0.28	1.12 ± 0.23
Healthy Adolescents	1.45 ± 0.40	113 ± 17.70	1.52 ± 0.25	1.09 ± 0.17
ACLD	1.24 ± 0.40	104 ± 18.30	1.39 ± 0.27	1.19 ± 0.21
ACLR	1.23 ± 0.38	102 ± 16.70	1.41 ± 0.24	1.21 ± 0.20

신뢰도 (CV): Gait speed 2.29–5.85%, Cadence 2.00–5.02%, Stride length 1.70–5.35%

P10 · Hur 2025

제목: Learning based lower limb joint kinematic estimation using open source IMU data 저자: Benjamin Hur, Sunin Baek, Inseung Kang, Daekyum Kim 저널: Scientific Reports (Nature Portfolio) 연도: 2025 · DOI: 10.1038/s41598-025-89716-4 파일: [Hur_2025_learning_lower_limb_kinematics_IMU.pdf](#)

Input Variables

IMU 원시 신호 (8개 센서 × 9채널 = 72 variables)

원문 발췌 — 입력 구조 (Deep learning, p. 4):

"we processed the data from 8 IMUs to have 72 variables for each time step as a deep learning input, encompassing x, y, z measurements of acceleration, gyroscope, and Euler angles."

채널	측정값	축
Acceleration	선형 가속도	x, y, z
Gyroscope	각속도	x, y, z
Euler angles	방향 (orientation)	x, y, z

- 총 입력: 9 channels × 8 IMUs = 72 variables
- 슬라이딩 윈도우: 50 samples (0.5초)
- 입력 텐서: [72 × 50]

원문 발췌 — 전처리 (Data preprocessing, p. 3):

"We normalized the accelerations, the angular velocities, and the Euler angles, to have zero mean and unit variance."

IMU 부착 위치 (8개)

원문 발췌 — 센서 위치 (Fig.2 캡션, p. 4):

"Total of eight IMUs, positioned on the torso, pelvis, bilateral femurs, tibias, and calcanei, are used as model inputs."

위치	수
Torso (흉부)	1
Pelvis (골반)	1
Bilateral femurs (양측 대퇴)	2
Bilateral tibias (양측 경골)	2
Bilateral calcanei (양측 발꿈치)	2

Target Variables (예측 대상 관절각, 10개)

원문 발췌 — 출력 변수 (Fig.2 캡션, p. 4):

"The model outputs are the bilateral sagittal, frontal, and transverse hip, sagittal knee, and sagittal ankle joint angles."

관절	면	양측
Hip	Sagittal (굴곡/신전)	좌/우
Hip	Frontal (내/외전)	좌/우
Hip	Transverse (내/외회전)	좌/우
Knee	Sagittal (굴곡/신전)	좌/우
Ankle	Sagittal (배굴/저굴)	좌/우

결론 및 주요 Feature

원문 발췌 — LSTM vs IMU-IK 비교 (Results, p. 5):

"The UI outperformed IMU-based inverse kinematics in RMSE values, NRMSE values, and Pearson correlation coefficient for both LSTM and CNN models. Specifically, the LSTM model showed 49.20% lower average RMSE, 50.65% lower average NRMSE, and 20.13% higher correlation coefficient compared to IMU-based inverse kinematics."

원문 발췌 — 핵심 IMU 조합 (Discussion, p. 8):

"femur and calcaneus are the most critical IMU placements, whereas the pelvis is the least important."

원문 발췌 — Transfer Learning 효과 (p. 6):

"applying transfer learning to a generalized pre-trained model improves estimation performance by 53.86%"

모델 성능 (Femur+Calcaneus 조합):

방법	평균 RMSE	특징
UI (개인별 학습)	Hip 3.83°, Knee 5.82°, Ankle 3.18°	최고 성능
UA (Transfer Learning)	Hip 14.03°, Knee 16.98°, Ankle 7.64°	UG 대비 53.86% 개선
IMU-IK	—	비교 기준

핵심 중요 변수: Femur + Calcaneus IMU 조합 (sagittal hip & knee 추정 핵심), Tibia (ankle 추정 보조); Pelvis는 중요도 낮음

제목: A Deep Learning-Based Framework Oriented to Pathological Gait Recognition with Inertial Sensors **저자:** Lucia Palazzo, Vladimiro Suglia, Sabrina Grieco, Domenico Buongiorno, Antonio Brunetti, Leonarda Carnimeo, Federica Amitrano, Armando Coccia, Gaetano Pagano, Giovanni D'Addio, Vitoantonio Bevilacqua **저널:** Sensors (MDPI), Vol. 25, No. 260 **연도:** 2025 · DOI: 10.3390/s25010260 **파일:** Palazzo_2025_deep_learning_pathological_gait_recognition_IMU.pdf

Input Variables

IMU 원시 신호 (5개 센서 × 9채널 = 45 channels)

원문 발췌 — 입력 구조 (Preprocessing, p. 6):

"Data from the inertial sensors were initially acquired for each subject and repetition as a matrix of size $N_s \times (5 \cdot 9)$, where N_s is the number of samples in a single repetition, 5 is the total number of IMU sensors, and 9 is the total number of IMU components (3 for the accelerometer, 3 for the gyroscope, 3 for the magnetometer)."

채널	측정값	축
Accelerometer	선형 가속도	x, y, z
Gyroscope	각속도	x, y, z
Magnetometer	자기장 세기	x, y, z

- 총 채널: $9 \times 5 = 45$ channels
- 샘플링: 128 Hz
- 윈도우: 128 samples (1초), 50% overlap

IMU 부착 위치 (5개)

원문 발췌 — 센서 위치 (Section 3.1.3, p. 5):

"Five sensors were selected (see Figure 1) and worn by each participant on both sides of the human pelvis (RP and LP), on the right and left wrists (RW and LW), and on the sternum (S)."

센서 코드	위치
S	Sternum (흉골)
LP	Left Pelvis (좌측 골반)
RP	Right Pelvis (우측 골반)
LW	Left Wrist (좌측 손목)
RW	Right Wrist (우측 손목)

측정 장비 사양 (Section 3.1.3, p. 5):

"The experimental data were recorded with a sampling rate of 128 Hz from a 3-axis 14-bit accelerometer to measure linear acceleration, a 3-axis 16-bit gyroscope to acquire angular velocity, and a 3-axis 16-bit magnetometer for magnetic field intensity."

Target Variables (분류 클래스)

원문 발췌 — 분류 대상 (Section 3.1.2, p. 4):

"in addition to normal walking, four pathological gaits were considered and they are ataxic, equine (foot drop), hemiplegic, and Parkinsonian gaits."

1. Normal gait (정상)
2. Hemiplegic gait (편마비)
3. Equine gait / Foot Drop (족하수)

- 4. Ataxic-cerebellar gait (소뇌성 실조)
- 5. Parkinsonian gait (파킨슨)

결론 및 주요 Feature

분류 성능 (제안 프레임워크):

보행 클래스	Accuracy
Normal	100%
Hemiplegia	100%
Ataxia	100%
Foot Drop	100%
Parkinsonian	100%

원문 발췌 — 성능 결과 (Section 4, p. 9):

"The median of accuracy and recall for the two models is 100% in almost all sensor configurations, despite the dataset imbalance."

원문 발췌 — 센서 성분별 비교 (Section 4, p. 10–11):

"The outcomes show better performance in both accuracy and recall for the accelerometer and gyroscope, whereas the outcome worsens when passing the magnetometer entry as input to the model, since the recall is about 80% only in three combinations."

원문 발췌 — Wrist 센서 중요성 (Section 4, p. 11):

"the motor behavior of human wrists is different for each type of walking action: it is stationary on the affected side in the hemiplegic gait close to the sternum, and it follows hand tremors in the Parkinsonian gait and arm sway in normal walking."

핵심 중요 변수: **Accelerometer + Gyroscope** 조합 (최고 성능); Magnetometer 단독 시 recall 약 80%로 저하; **Wrist 센서** — 병리 보행 유형 변별에 핵심 역할

전체 Input Variables 통합 비교표

변수 카테고리	구체 변수	사용 논문
vGRF	Peak vertical GRF, GRF asymmetry	P1, P4, P5
PA-GRF	Posterior-anterior GRF, asymmetry	P5
Knee kinetics (sagittal)	Knee flexion moment (KFM), knee extension moment (KEM1/2)	P1, P3, P4
Knee kinetics (frontal)	Knee adduction moment (KAM1/2, impulse)	P1, P3
Knee kinetics (transverse)	Internal/external rotation moment (KIRM/KERM)	P3
Hip kinetics	Hip adduction/abduction moment	P1
Knee kinematics	Knee flexion angle (IC, peak, stance), knee abduction/varus/valgus angle	P1, P3, P4
Hip kinematics	Hip abduction angle, external rotation	P1, P3
Ankle/foot kinematics	Ankle dorsiflexion/plantarflexion (0%, 25%, 60%, 83% GC), foot progression angle	P2

변수 카테고리	구체 변수	사용 논문
Spatiotemporal — gait speed	Walking speed, self-selected speed conditions	P1, P3, P5, P8, P9
Spatiotemporal — cadence	Steps/min	P7, P8, P9
Spatiotemporal — stride/step	Stride length, step length, step width, stride time	P3, P8, P9
Spatiotemporal — stance	Single/double leg stance %, double stance time	P2, P8
IMU: Acceleration (raw)	Acc_X/Y/Z (3축), free acceleration	P7, P8, P9, P10, P11
IMU: Angular velocity (raw)	Gyr_X/Y/Z (3축)	P7, P8, P9, P10, P11
IMU: Magnetometer	Magnetic field 3축	P11
IMU: Euler/Quaternion	Segment orientation	P9, P10
IMU: Time-domain features	Mean, STD, RMS, skewness, kurtosis, zero-crossing rate	P7
IMU: Frequency-domain features	Dominant frequency, spectral energy, spectral entropy	P7
Clinical — muscle strength	60°/s knee extensor/flexor PT, 180°/s knee extensor/flexor AP (isokinetic)	P6
Clinical — balance (static)	BBS overall index (anteroposterior + mediolateral)	P6
Clinical — balance (dynamic)	YBT total score (anterior, posteromedial, posterolateral)	P6
Demographics	Age, BMI, sex, height, weight	P6, P8, P9
Clinical scores	IKDC, WOMAC, UPDRS III, FMA-LE, TUG, visualGaitEvaluation	P8
Limb symmetry	GRF asymmetry (injured – uninjured), limb symmetry values	P4, P5
Anterior femoral displacement	Average during stance phase	P3

중요 Feature 요약

결론/Discussion/Feature Importance 기반으로 각 논문이 강조한 핵심 변수를 정리합니다.

논문	연구 목적	핵심 중요 변수	근거
P1 · Goetschius 2018	ACLР 시기별 보행 차이	vGRF, knee extension moment, knee adduction moment	Early ACLR: sagittal loading ↓ ; Late ACLR: frontal loading ↑
P2 · Ursei 2020	ACL 결핍 아동 발목 보상	Ankle dorsiflexion at 0% GC (initial contact)	$p=2.55 \times 10^{-9}$, 가장 유의한 ACL 결핍 지표
P3 · Erhart-Hledik 2018	ACLР 후 2→8년 종단 변화	KFM ↑, KEM ↓, External rotation ↑	$p<0.001$, OA 진행과 연관

논문	연구 목적	핵심 중요 변수	근거
P4 · Krishnan 2022	knee extension moment 기여 역학	Peak knee flexion angle ($\beta=0.703-0.831$)	vGRF 대비 2배 이상 영향력; $R^2=0.77-0.88$
P5 · Garcia 2022	보행 속도별 GRF 비대칭	vGRF asymmetry , PA-GRF asymmetry	속도 증가 → 비대칭 악화 (ACL 군만)
P6 · Hwang 2024	ML로 RTS 예측	YBT total score, 60°/s knee extensor PT, 180°/s knee extensor AP	SHAP 분석 기반; RF AUC=0.952
P7 · Yuan 2026	IMU 최적 구성 + ML	Accel_Z_variance (Lumbar) (Gini=0.187), Accel_X_variance	$R^2=0.99$ (Cadence); C3 (3 IMU) 비대칭 예측 필수
P8 · Voisard 2025	다병리 임상 IMU 데이터셋	Gait velocity, stride time, double stance time	정상 vs 병리 집단 변별
P9 · Yona 2026	ACL 다속도 IMU 데이터셋	Gait speed, cadence, stride length	ACL D/ACL R: gait speed 유의하게 낮음 ($p<0.001$)
P10 · Hur 2025	IMU→관절각 추정 (DL)	Femur + Calcaneus IMU (sagittal hip & knee); Tibia (ankle)	Pelvis 중요도 최저; Transfer Learning 53.86% 개선
P11 · Palazzo 2025	병리 보행 분류 (DL)	Accelerometer + Gyroscope , Wrist 센서	Magnetometer 단독 recall 80%로 저하; 모든 병리 100% 분류

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